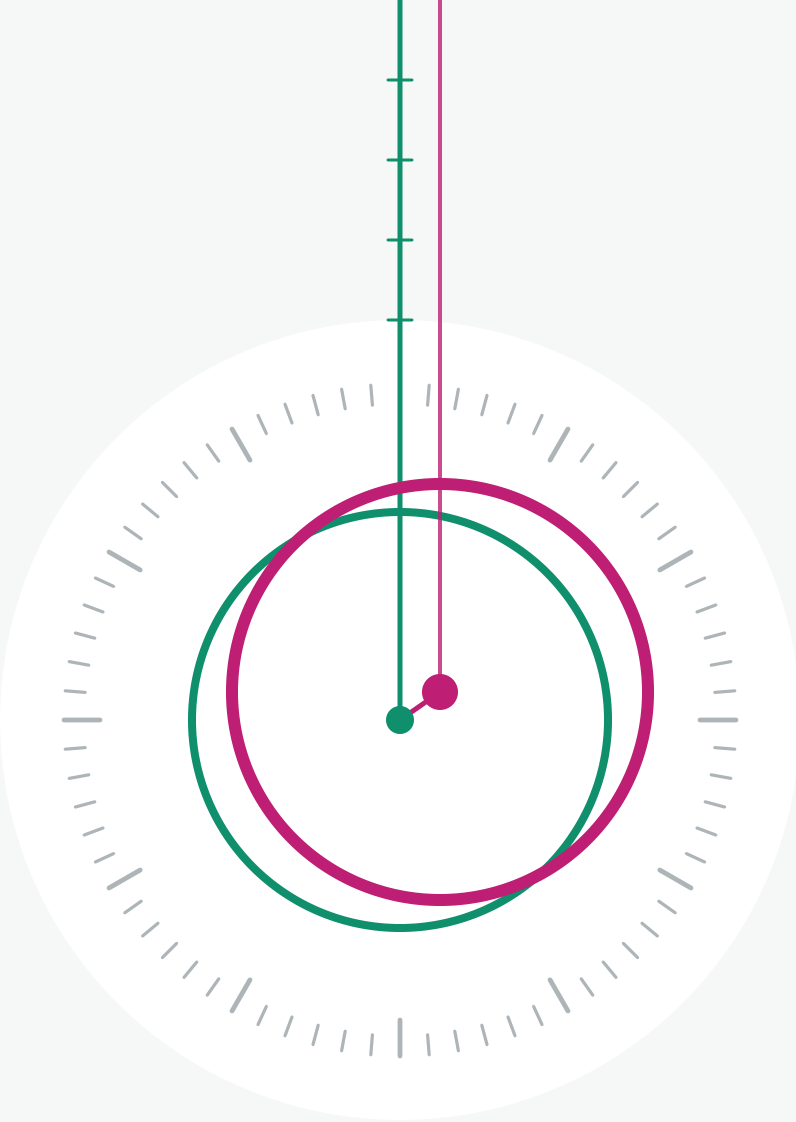
DATUM LINE

Confidence with controls.

Every lab’s cells, on one map.



Kat Titterton · San Francisco



Katherine Titterton

five labs · one **unanswered question**

2023–2026 **Axiom**
founding MTS · hepatocytes at scale

2022–2023 **insitro**
neuroscience assays · productionization

2021–2022 **MIT** SYNTHETIC NEUROBIOLOGY
expansion microscopy

2018–2021 **abbvie** NEUROSCIENCE
tau · neurons

2017–2018 **MGH** HARVARD MEDICAL SCHOOL
Xenopus oocytes · $GABA_A$ receptors

2017 **COLGATE** BA
neuroscience + math biology

macrofluidic.me/resume

ROLE

not the CEO
I'd build the science

looking for its leader
and its home

MY WORK

1. Cell Painting in primary human hepatocytes
   Cell Systems 2026
2. Counting cells predicts bioactivity benchmarks
   Nature Communications 2026
3. A neuronal tau aggregation assay
   J Biol Chem 2025
4. Multiplexed imaging of brain nanostructures
   Nature Communications 2024
5. Neurosteroid photolabeling of $GABA_A$ receptors
   Eur J Med Chem 2019

PUBLISHED WITH

Broad Institute · MIT · AbbVie · EMBL-EBI
HESI · Bayer · Cambridge · Uppsala

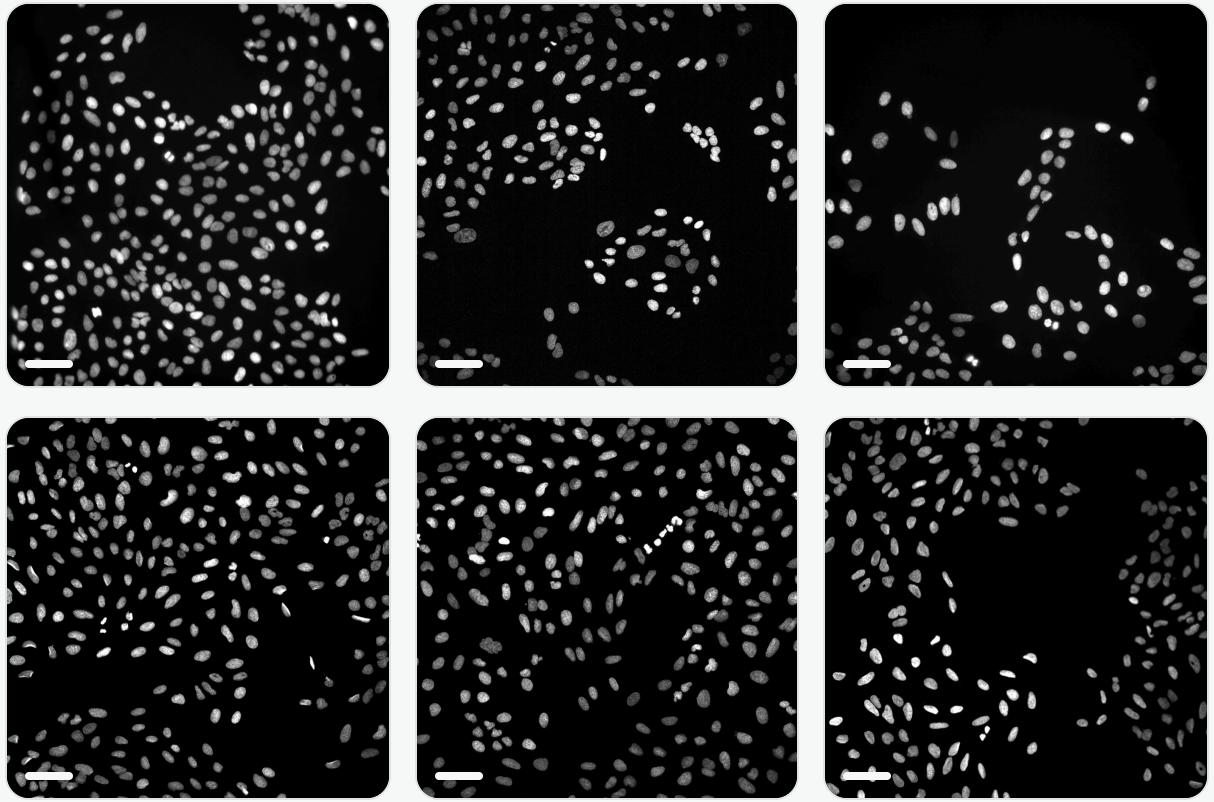
click → the question

Do my cells look normal?

asked at five benches
answered at none

↓ the game

One lab, or six?



same cells ✓

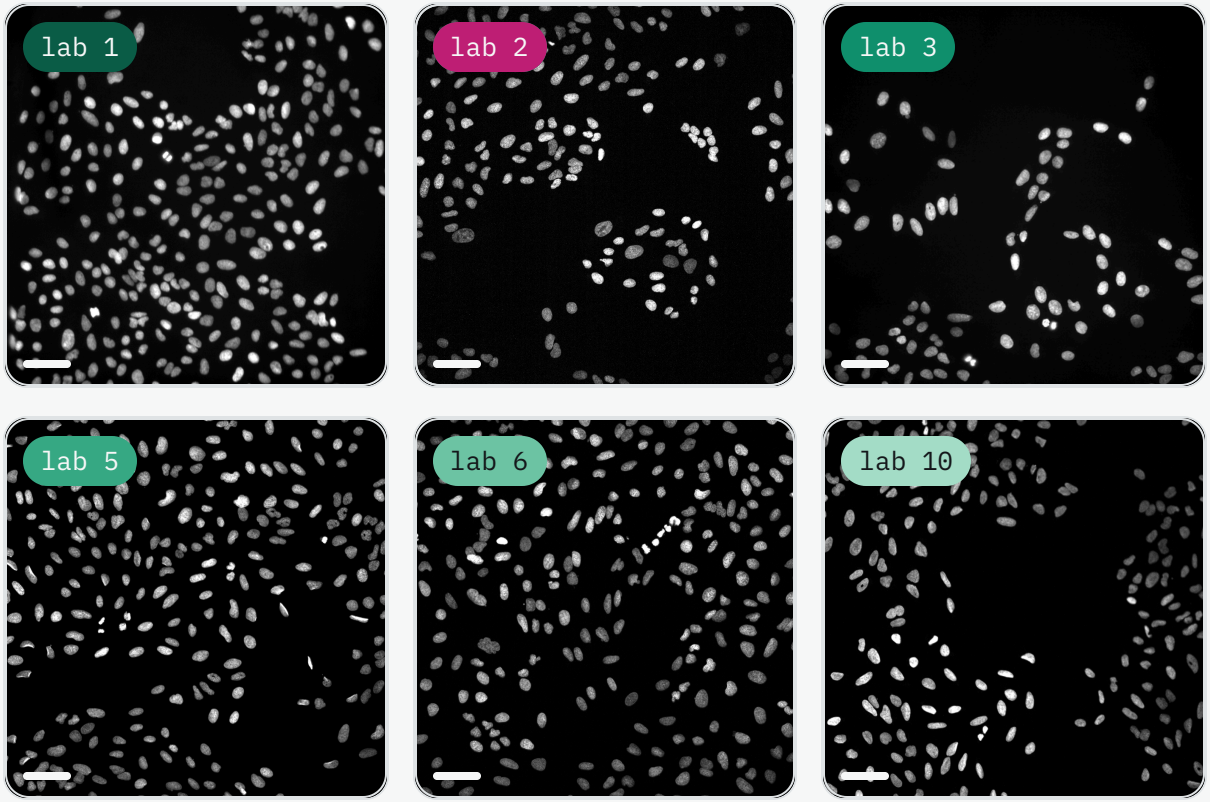
same protocol ✓

no drug ✓

fig 1 Cell Painting · DNA channel · 93,228 wells · 11 labs · plates held out · bars 100 µm · JUMP [7]

click → the answer

Could you tell?
Classifiers can.



99.9% CLASSIFIER

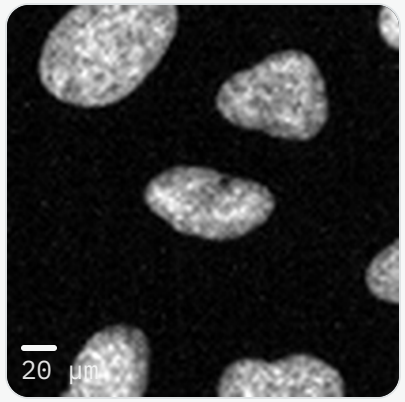
9% CHANCE · 1 IN 11 LABS

fig 1 Cell Painting · DNA channel · 93,228 wells · 11 labs · plates held out · bars 100 µm · JUMP [7]

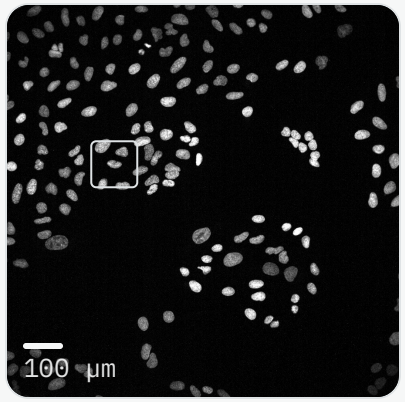
click → onto the map

One well, onto the map.

ONE NUCLEUS



ITS WELL



lab 2

3,773 numbers

one point

WHERE IT LANDS



fig 1 Cell Painting · DNA channel · 93,228 wells · 11 labs · plates held out · bars 100 µm · JUMP [7]

↓ the map

Eleven labs.
Eleven normals.



6.2

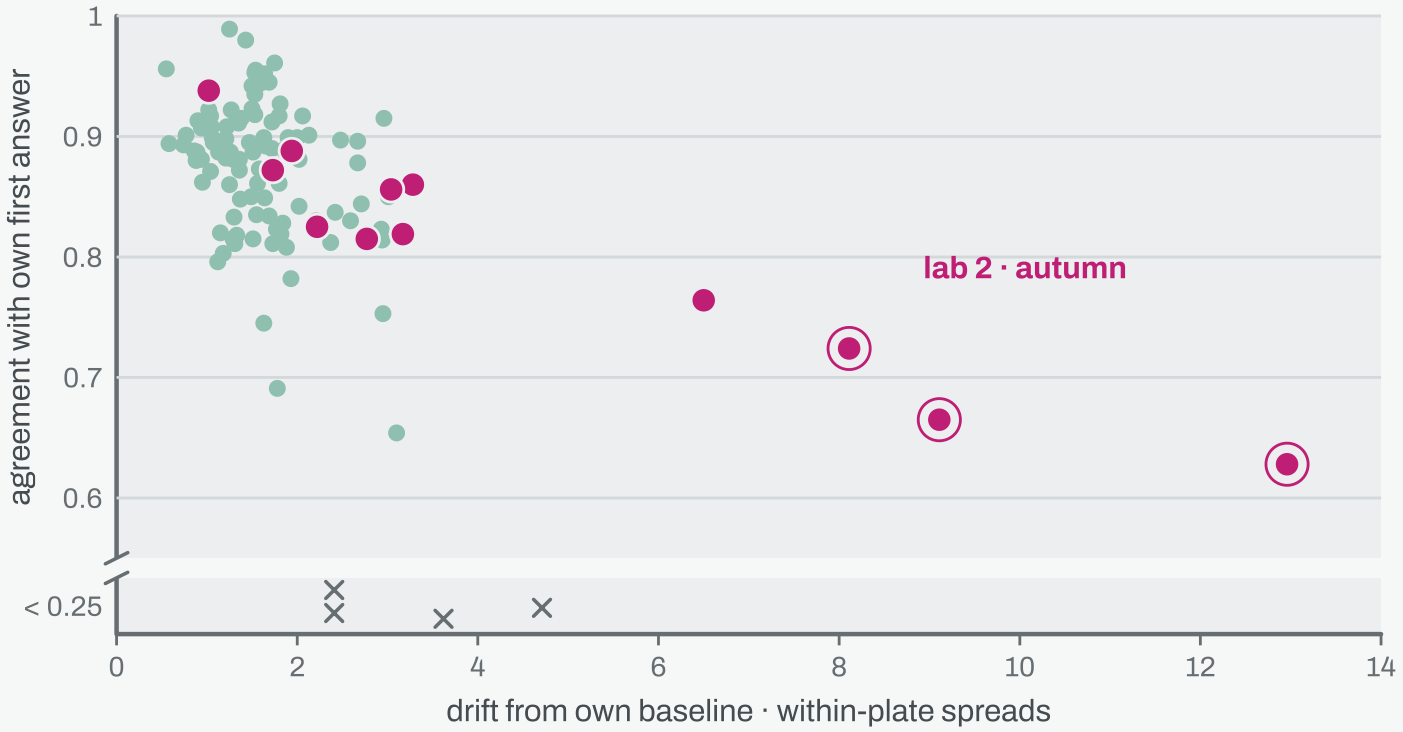
lab to lab

inside one plate

fig 2 15,352 untreated wells · PCA of 3,773 numbers · JUMP [7]

click → why the map matters

Where you sit predicts what you measure.



INSIDE LABS

ρ = −0.71

more drift · less agreement

lab 2 · same drugs

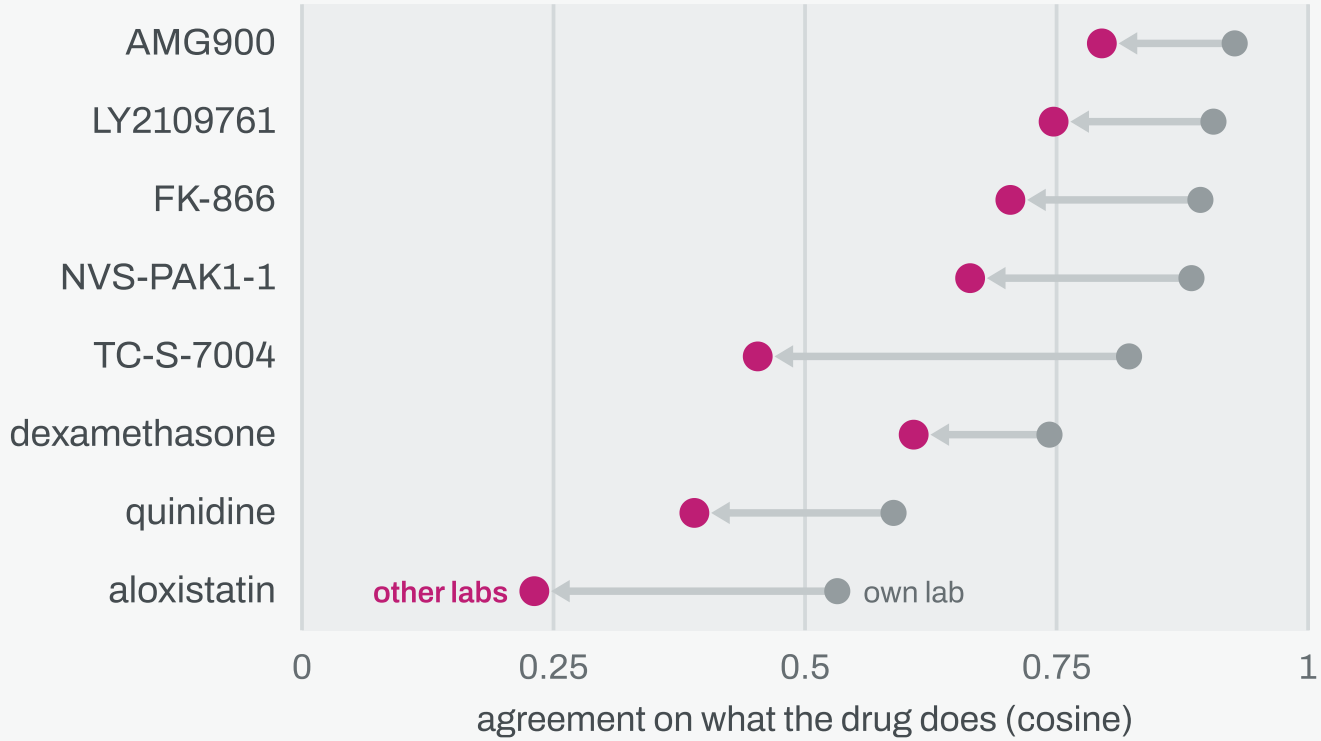
0.85 → 0.67

● → ◉

fig 3 118 batches · 11 labs · JUMP [7] · A8, A13

↓ positive controls

Does my control look like yours?



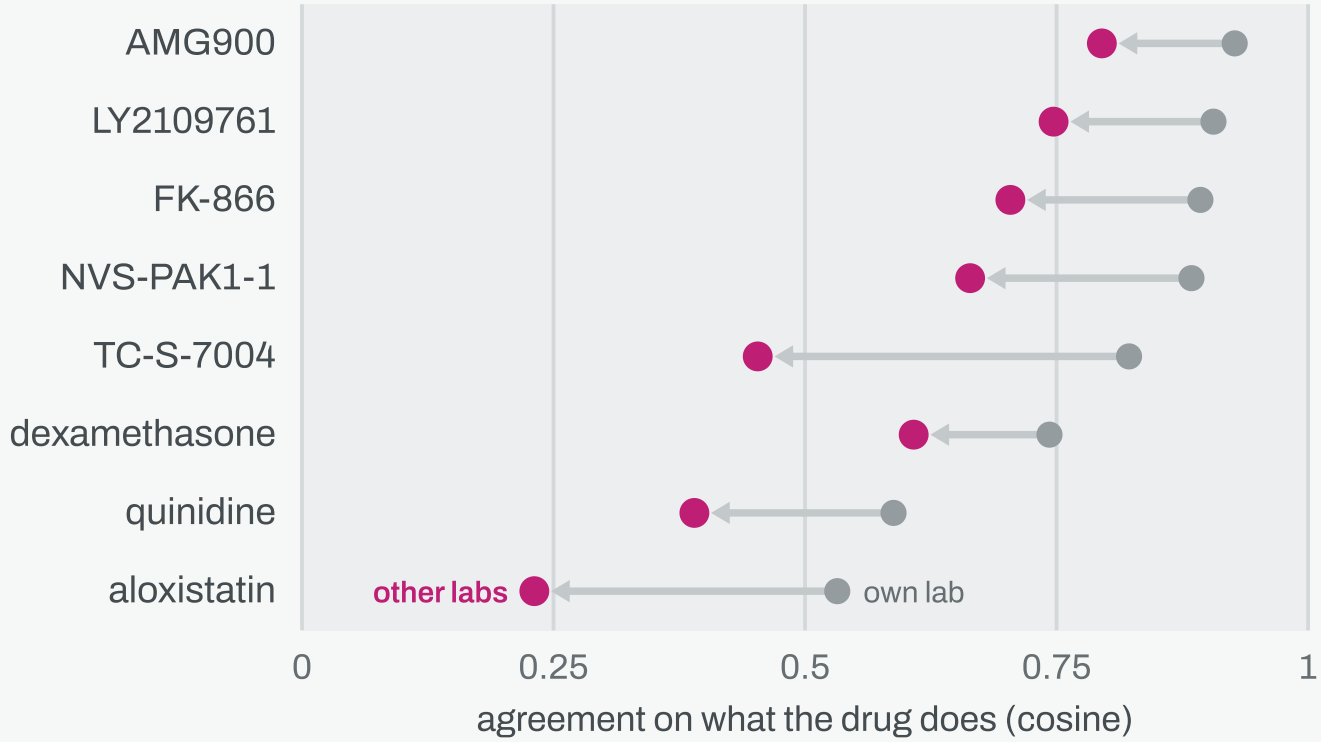
0.79 → 0.57

agreement per plate · own lab → other labs

fig 4 eight positive controls · 1,829 plates · 11 labs · JUMP [7] · A9, A14

click → on the map

Does my control look like yours?



0.79 → 0.57

agreement per plate · own lab → other labs

ON THE MAP · PER BATCH

0.75 → 0.78

agreement with other labs · up in 121 of 125

0.52 → 0.59

drugs told apart · up in every batch

Lab comes out.
Drug stays.

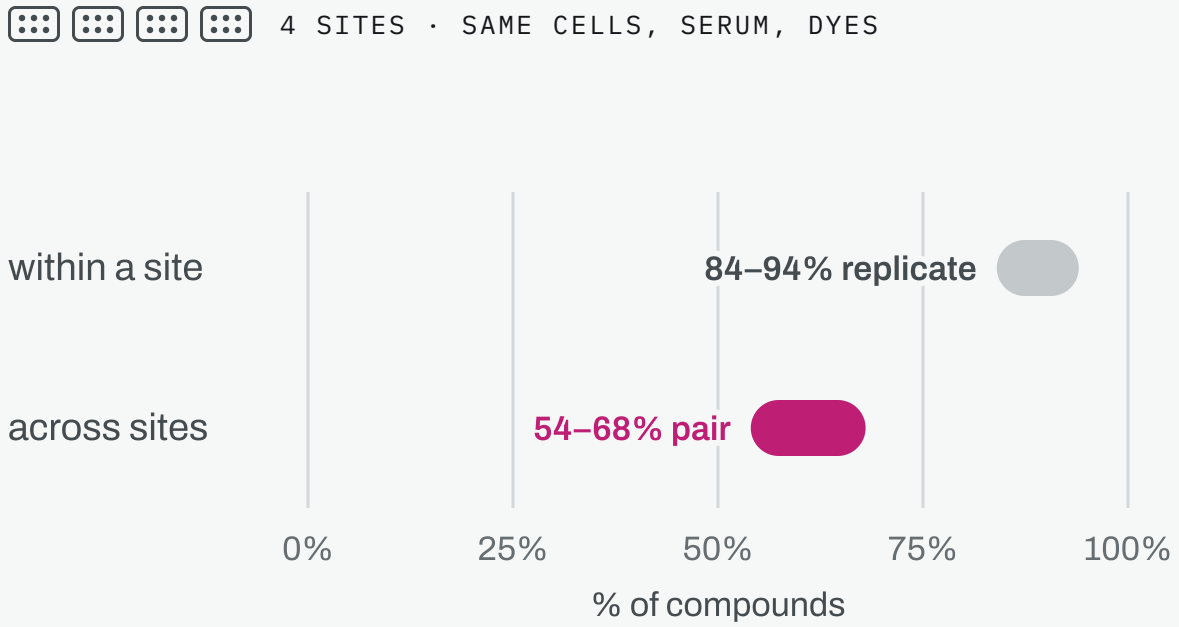
fig 4 eight positive controls · 1,829 plates · 11 labs · JUMP [7] · A9, A14

↓ the data economy

Data is the new currency.
Cell data doesn’t convert.

$30M

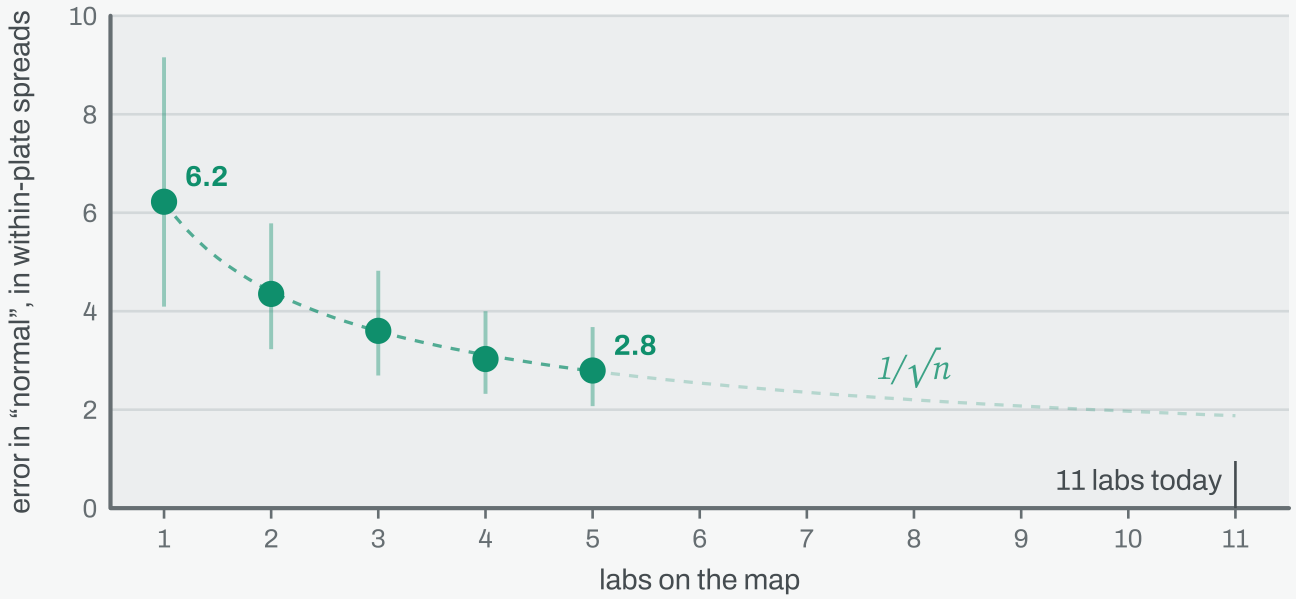
one cell map · paid by Roche [11]



“not directly comparable between imaging sites” [13]

fig 5 EU-OPENSCREEN · one protocol · Hep G2 · as reported [13] · A23

Every lab that joins sharpens the map.



THE MAP’S ERROR

6.2 → 2.8

one lab → five

MAP open — today → CERTIFICATE paid — next → EXCHANGE a share of each trade — then → …

fig 6 two maps, different labs · 1,871 plates · JUMP [7]

↓ the product

Drop an image.
Get an answer.



Your answer

LAB 2 · U2OS · 16 AUG

0.70 → 0.77

agreement with other labs · this batch

WHERE YOU SIT

in distribution

2.8 from your first batch
4.4 to lab 6 · nearest

YOUR OWN CHECK

188 cells per well · in range

fig 7 replayed on public data · JUMP [7] · A11

click → the leaderboard

A game you can win.

LEADERBOARD · U2OS · 129 BATCHES · 11 LABS

controls only

RANK	LAB	BATCH	PLATES	FIELD SCORE	/100
#1	lab 5	28 Jun	7 pl		82
#2	lab 5	2 Jul	6 pl		79
#3	lab 5	6 Jul	5 pl		79
#4	lab 5	16 Jul	8 pl		78
#5	lab 5	1 Oct	6 pl		77
…					
#48	lab 2	2 Aug	18 pl	your best	58
…					
#95	lab 2	16 Aug	25 pl	this batch	36

Your standing

LAB 2 · 16 AUG

#95 of 129 batches · score 36

YOUR STREAK

field best 33 · lab 5

TO CLIMB · DRIFT

you	2.9
field	1.2

fig 8 score = 100 − mean percentile · untreated wells · JUMP [7] · A17

click → the exchange

Your data, certified.
Ready to trade.

EXCHANGE · U2OS CONTROLS · 11 DATASETS

certificate attached

DATASET	PLATES	BATCHES CERTIFIED	CONVERTS
lab 3	237	■■■■■■□■■■□■■	0.80 → **0.81**
lab 6	225	■■■■■■■■■■■■■	0.71 → **0.76**
lab 2	219	■■■■■■■■■■□□□	0.75 → **0.80**
lab 10	205	■■■■■■■■■■■■■■■■■■■■	0.80 → **0.81**
lab 8	199	■■■■	0.84 → **0.88**
lab 5	197	■■■■■■■■■■■■■■■■■■■

Six questions, six tools.

TOOL	QUESTION IT ANSWERS	DOES NOT SEE
STR authentication	Is it the line the label says?	behaviour
vendor certificate	Was it fine when it shipped?	anything after shipping
plate controls	Did this plate work?	a whole lab drifting
phenomics platforms	Is my own data consistent?	lab differences, by design
data factories	Where do I get cell data at scale?	other labs' normal
Datum Line	**Where do my cells sit, against every other lab?**	which line · beyond fluorescence, yet

use all of them · A3, A7, A23

Eighteen months, four milestones.

$1.8M to build it · A21

1

three labs on the map

U2OS first · any line

2

model for any image

99.6% from one brightfield image · A19

3

benchmark playground

cell lines · microscopes · perturbations · A16

4

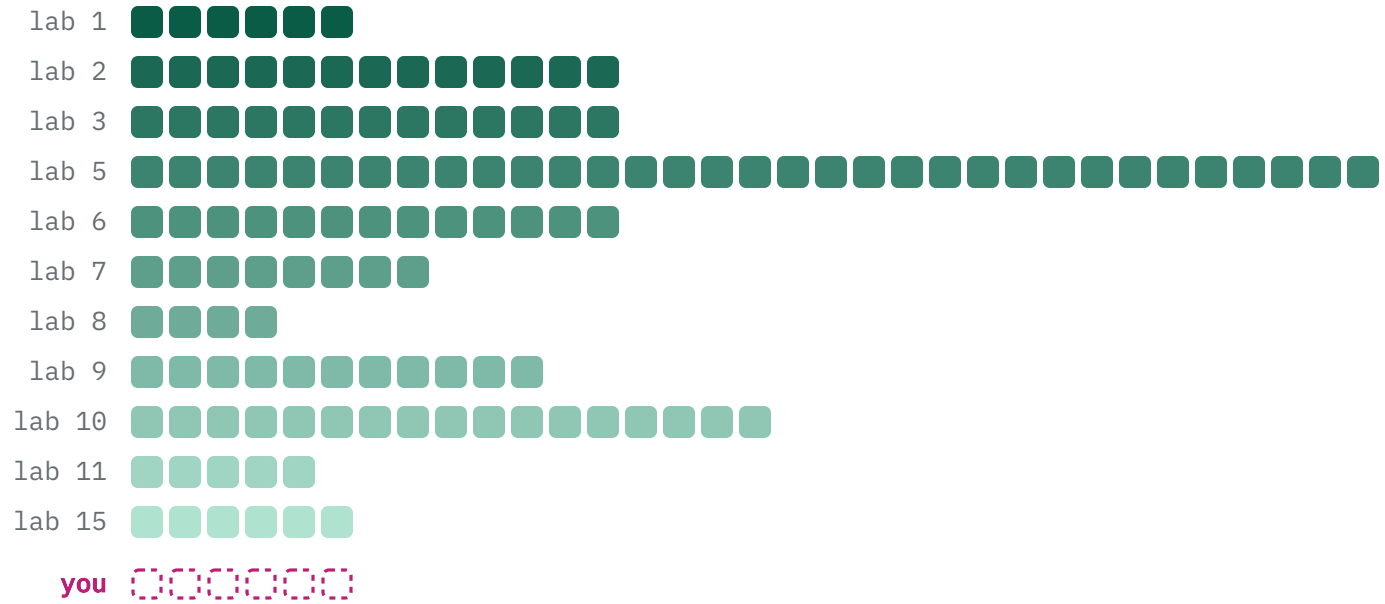
first paid certificate

on data a company bought

then → exchange

proposed · A23, A24

Every plate you image, placed on the map.



MAP
today
→
CERTIFICATE
next
→
EXCHANGE
then
→
…

WHAT IT NEEDS

- a leader
- images
- bench
- funding
- partners

Have control images? Talk to me.

fig 10 one square per batch · 129 batches · 1,871 plates · 11 labs · JUMP [7]

The rest, if you ask.

Every number traces to results/ or a numbered reference.
Pending work says pending.

A1 **Why not trust the plate's controls?**
there is no control well

A2 **What is the math?**
six operations · one vector

A3 **Why not batch-correct it away?**
what JUMP records

A4 **Is the model the moat?**
counting cells often wins

A5 **Why not just count cells?**
same count · different texture

A6 **How often do I re-certify?**
line is not expired

A7 **Isn't this STR?**
STR says who · images say how

A8 **Is the assay itself working?**
known answer

A9 **Does it change the result?**
same drug · different answer

A10 **Does it work without stains?**
bet on brightfield

A11 **Would it cry wolf?**
six of 129 batches

A12 **What moved?**
stain intensity, not cell count

A13 **How does one lab move over time?**
lab's check · field's view

A14 **What does the field's frame buy?**
agreement between labs

A15 **Who pays?**
labs free · buyers of cell data pay

A16 **What would the money build?**
ugliest dataset · on purpose

A17 **Is the leaderboard fair?**
controls only, never results

A18 **What is private?**
control wells leave · nothing else does

A19 **What is the machine learning?**
image in · place, distance, move out

A20 **What about biosecurity?**
cells cannot lie about their origin

A21 **How much, and what for?**
planned at $0.9M · asked at 2x

A22 **How big?**
count × price · three streams

A23 **What is cell data worth?**
priced in deals · sites do not add up

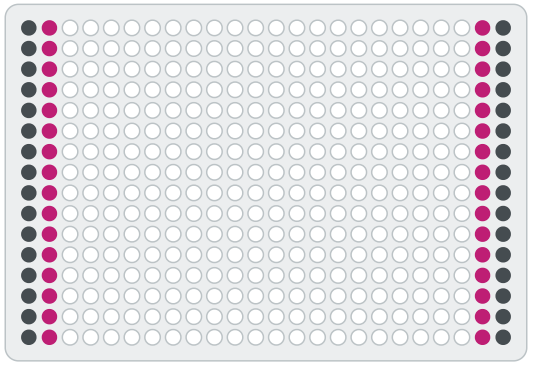
A24 **Why would a lab share?**
owned by the lab · verified by anyone

A25 **How do labs answer it today?**
search, ask, guess

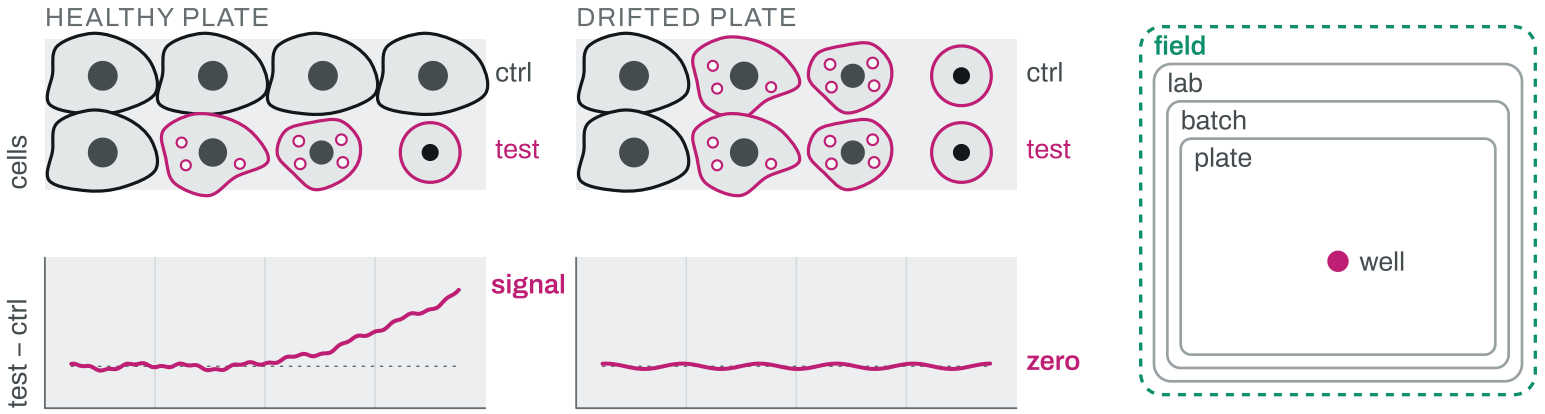
There is no control well.

A control only sees inside its own box.

A REAL JUMP PLATE · 384 WELLS



DMSO 32 · positive 32 · test 320



$$x_{test} = \mu + s + \tau(s)$$

$$x_{ctrl} = \mu + s$$

$$\Delta = \tau(s)$$

s · the plate's state · cancels from the number, never from the cells

If the plate is sick, so is the control.
Then how would anyone know?

Fig. A1 · left: JUMP plate 1053601756, lab 2 · right: schematic, no measurements

Six operations, one vector.

Every well: $e \in \mathbb{R}^{3773}$, z-scored across the field.
Each result: a difference of means or a held-out classifier.

1 · EMBEDDING [1]

$$e = \varphi(image)$$

ϕ ∈ {CellProfiler, CNN, ViT}: similar on hepatocytes.
Extractor is not the asset.

2 · LAB CLASSIFIER 99.9%

$$f : e \to lab$$

L2 logistic regression, five folds held out by plate.
Chance 1 in 11 · permuted labels 9.3%
Without 1536-well labs: 99.98%

3 · DISTANCE, THE CERTIFICATE [e6]

$$d = \| E[e \mid plate] - E[e \mid field,\ c] \|_{\sigma}$$

Difference of means, in units of the field's spread.
Its direction is what moved; −d is the way back.

4 · VARIANCE DECOMPOSITION pending

$$Var(e) = V_{model} + V_{lab} + V_{batch} + V_{plate} + V_{well}$$

Nested: shares sum to one.
Microscope models span labs, so the two separate.

5 · REGRESS VARIABLES OUT 24%

$$e_{\perp} = e - \hat{E}[e \mid c]$$

c: known instrument variables only; then re-run f.
Plate-centred alone: f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS

DISTANCE $\delta = f(\|e - E[e \mid vehicle]\|)$

CLASSIFIER $g : e' \to label$

f bounded · e′ aligned to the reference first

discovery · align the lab away first
QC · keep the lab coordinate, never align it away

results/e1, e1-redteam, e1b, e6 · e5 pending · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is *E[x | every variable below]*.

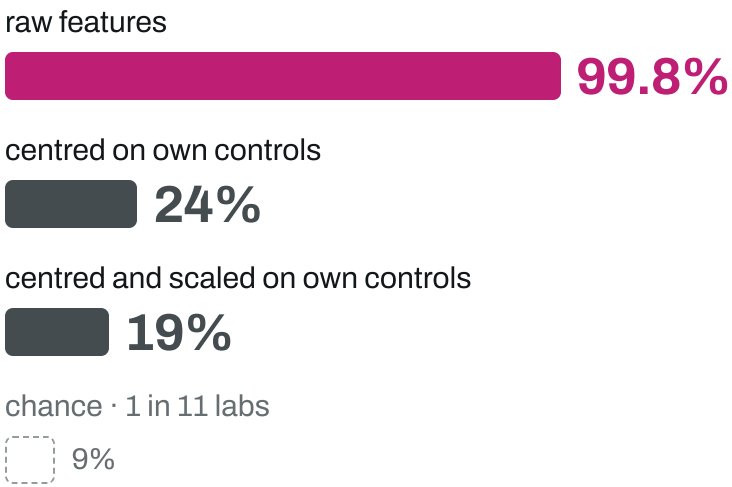
RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NAMING THE LAB · BALANCED ACCURACY



29,943 wells · 16 per plate · plates held out

Centring each plate on its own controls is the standard first step.
It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo et al., Nat Commun 2024 · variables: JUMP metadata [7]

NATURE COMMUNICATIONS · 2026 · [2]

Weak benchmarks: counting cells often wins.

I coauthored this paper.

… Unfortunately, we discovered that many assays in widely used assay-activity benchmark datasets directly relate to cell health and cytotoxicity. Many other assays intend to capture a more specific phenotype, but their active compounds impact cell count, while inactives do not. In both cases, counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. To accomplish this goal, we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. Using a benchmark with 24 protein-target assays, we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. …

ABSTRACT EXCERPT · VERBATIM `finding` `recommendation · what held up`

209 and 270

assays in two widely used benchmarks
prior work: AUC above 0.9 on 32% of 209

24

protein-target assays
on them, Cell Painting profiles beat the count

WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark.

Datum Line benchmark · cell-count baseline on every task · planted failures · every model graded · A16

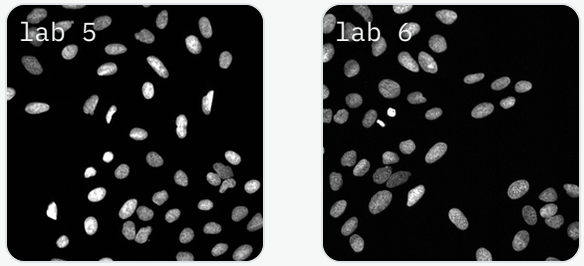
Seal S, Dee W, … **Titterton K**, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

CELL SYSTEMS · 2026 · [1]

Same count.
Different texture.

Count moves only when cells die or divide.
Image moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



196	≈	222	count
0.94	≠	1.10	texture

METHOD 1 · CLASSIFIER

Does this run belong with the past runs?

$g: e' \rightarrow belongs / does\ not,\ \ e' = align(e)$

standard alignment · our tuning



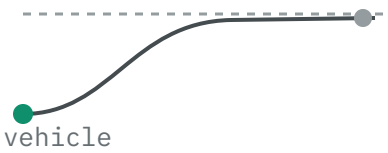
cells, not the day

METHOD 2 · DISTANCE

How far out, and which way back?

$\delta = f(\|e - E[e \mid vehicle]\|),\ \ f\ bounded$

distance published [1] · transform unpublished



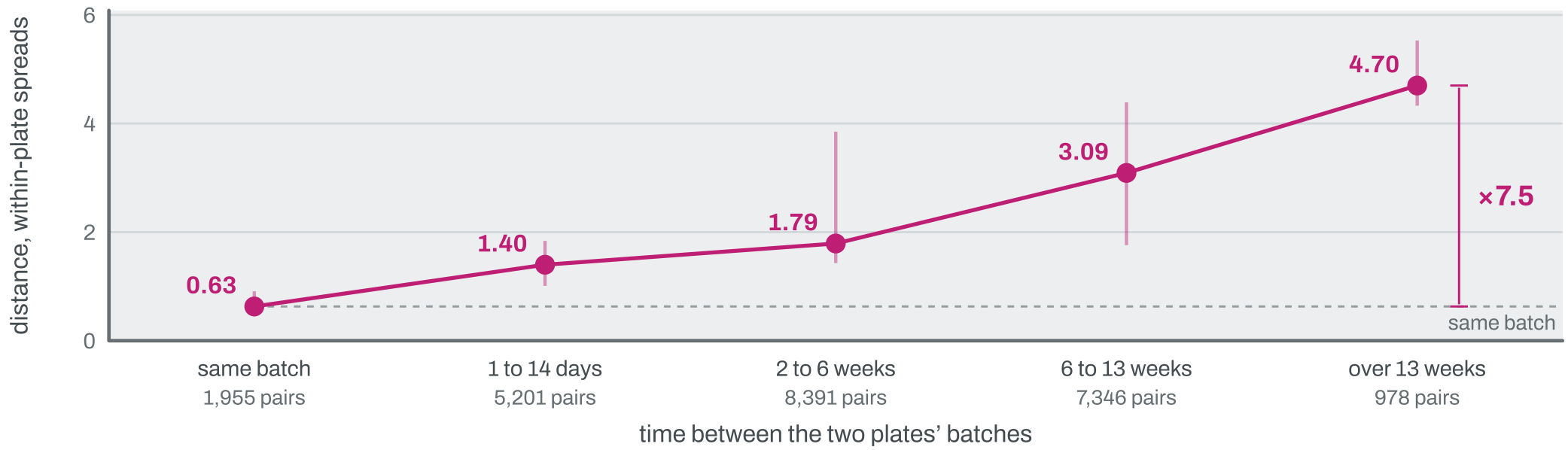
one wild well can’t swamp it

both need a reference · somebody else’s images

Fig. A5 DNA channel · JUMP cpg0016 [7], CC0 · nuclei by Otsu · texture = high-frequency energy over total, inside nuclei · results/a5-pair

The line is not expired.
Your baseline is.

Plates of one lab drift apart with time.
Still rising after 13 weeks.



IS THE CELL LINE RUINED?

Unknown from profiles.
Data shows when the lab moved, not why.

HOW OFTEN?

At least every quarter.
Distance ×4.9 by 6 to 13 weeks.

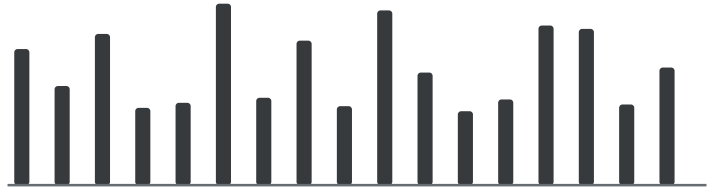
HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches.
Other labs not yet measured.

Median distance between plate centroids, in within-plate spreads, with the interquartile range.

STR says who.
Images say how.

STR AUTHENTICATION

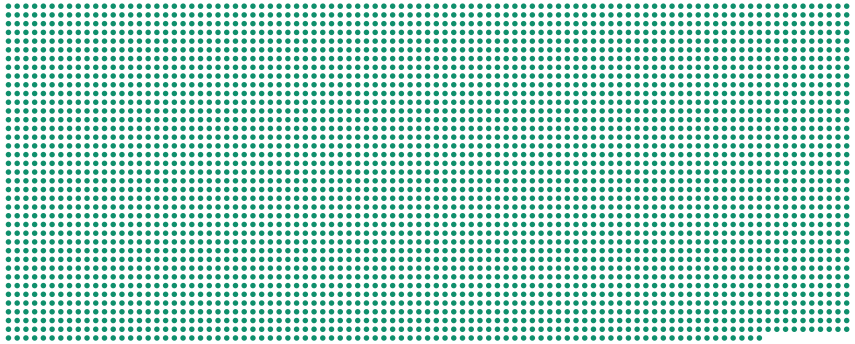


17
loci

once
per line · mailed card

Standard for identity.
Cannot say how it behaves.

DATUM LINE



3,773
numbers per well

every batch
images already taken

Behaviour on the field's map.
Cannot say which line it is.

identity · STR + behaviour · Datum Line = **one certificate**

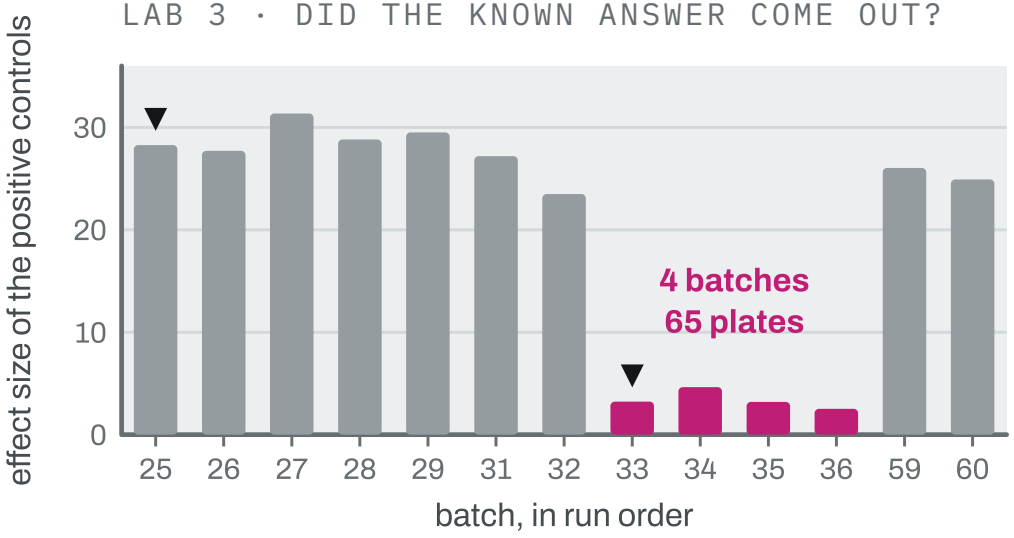
STR provider · natural first integration

STR: ATCC [14], 17 loci plus amelogenin. Datum Line: 93,228 untreated wells, 1,872 plates, 11 labs, JUMP [7].

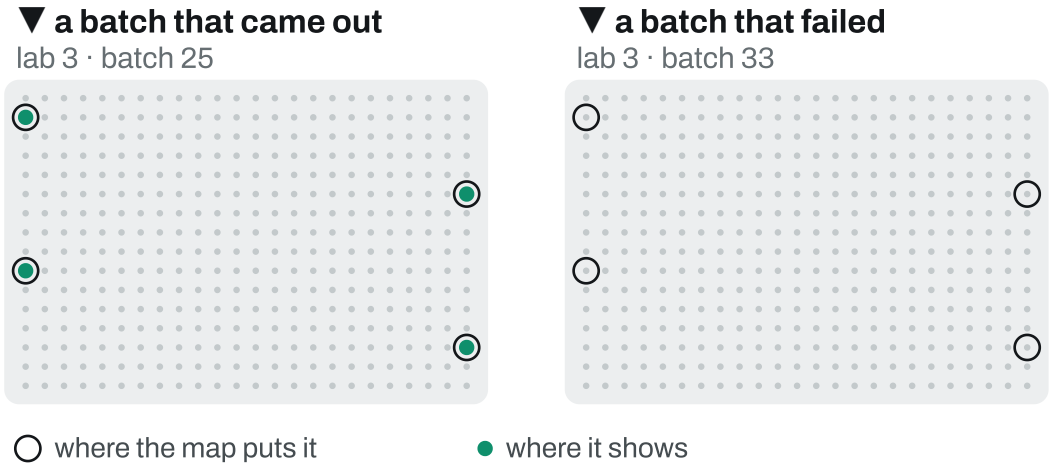
A known answer, run in every lab.

Untreated wells say where a lab sits.
Positive controls say whether it answers.

LAB 3 · DID THE KNOWN ANSWER COME OUT?



WHERE ON THE PLATE DOES AMG900 SHOW?



125 of 129

JUMP BATCHES · KNOWN ANSWER CAME OUT

1 of 4

FAILED BATCHES · FLAGGED BY UNTREATED WELLS

Agreement with the field: .03–.11 failed, .76–.83 the rest. 1,829 plates, JUMP [7]. Plates C13443aW, A12069aW: public JUMP plate map.

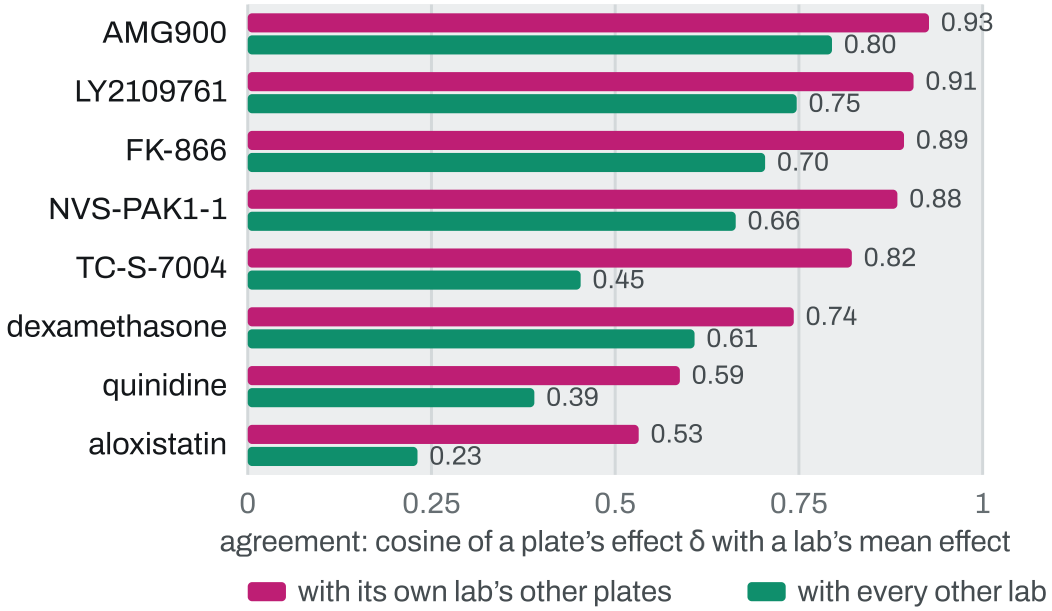
Same drug.
Different answer.

8 positive controls, 11 labs.
Between labs, each drug's effect moves in direction and size.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{compound } c] - E[x \mid \text{plate } p, \text{DMSO}]$$

HOW WELL A PLATE AGREES ON WHAT A DRUG DOES

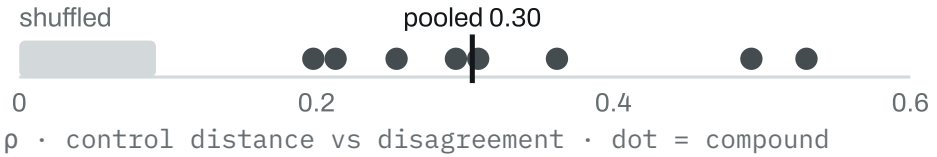


Q1 · DOES IT AGREE?

Less with other labs, for every compound.

Q2 · DOES IT TRACK THE CONTROLS?

Inside labs, yes.



Q3 · DOES IT MATTER?

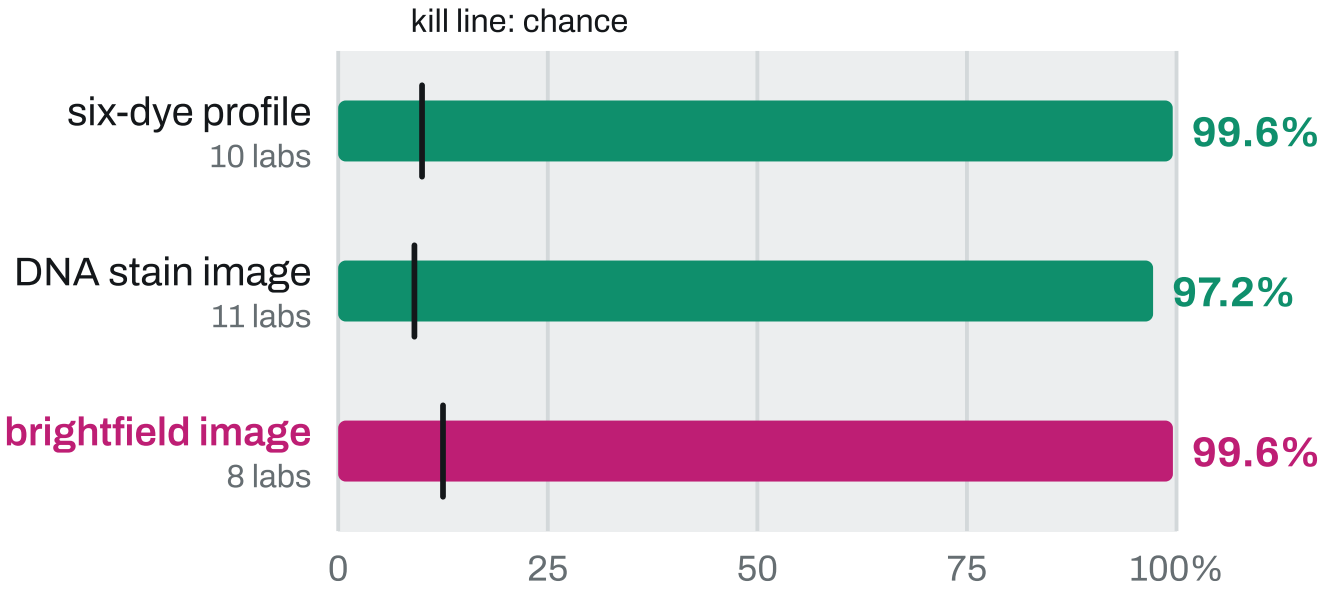
20% to 71% of effect-size variance is between labs.

8 positive controls · 1,829 plates · 3,757 features · 11 labs · JUMP [7] · z-scored on every other lab's DMSO wells, clipped at 5

Shown in fluorescence.
Bet on brightfield.

Evidence: six dyes.
Product: no dye.

NAMES THE LAB · WHOLE BATCHES HELD OUT



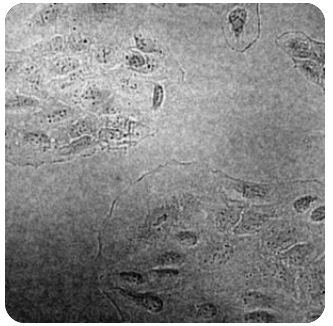
Next: the whole map from brightfield.

Today: nearest lab agrees with the profile map in 29 of 102 batches.

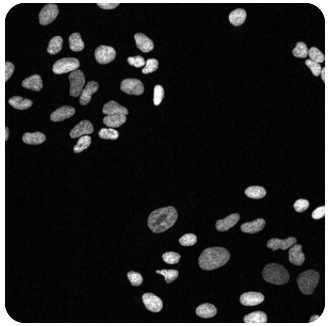
WHY THE BET IS FAIR

Fewer variables, not more

~~stain lot~~ · ~~dye lot~~ · ~~staining step~~ A3



BRIGHTFIELD



DNA STAIN

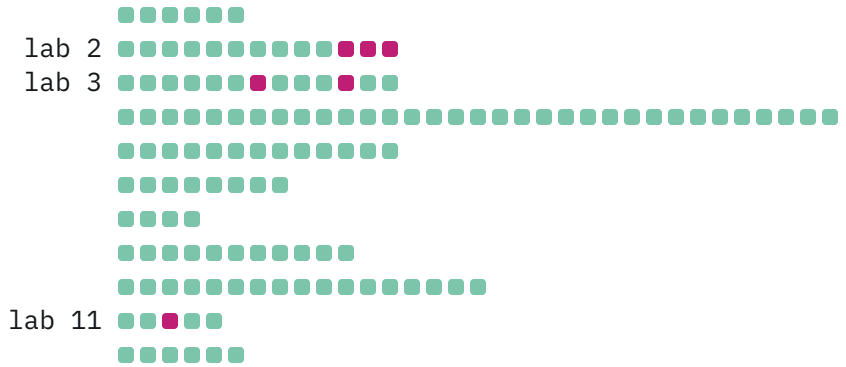
same field · lab 2 · June 2021

balanced accuracy · whole batches held out · images: one field of one untreated well, frozen vision model, no training on cells · JUMP [7]

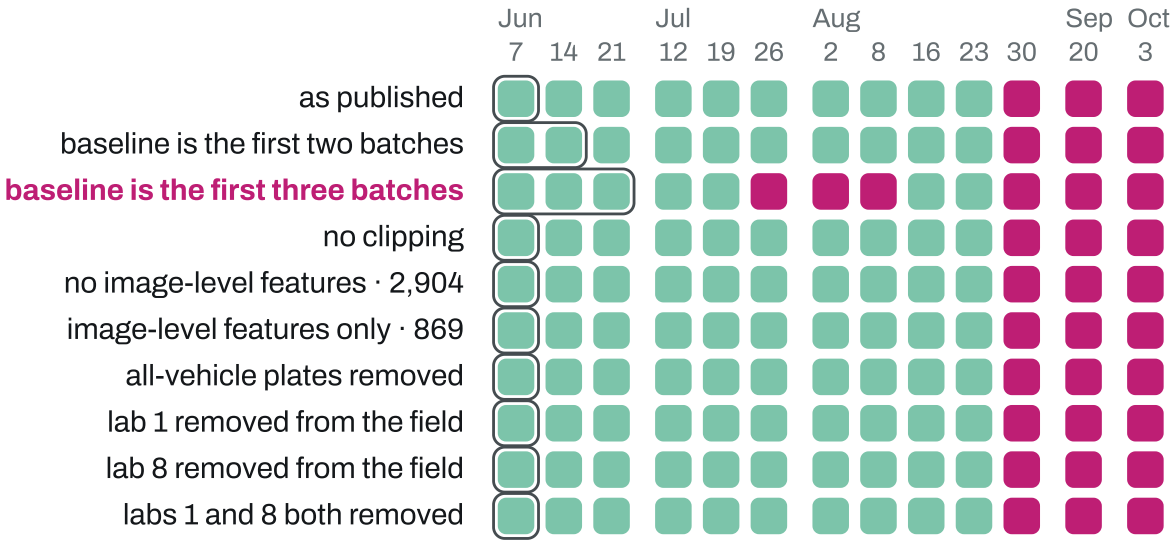
Would it cry wolf?
6 of 129.

Rare across the field.
Same verdict under 9 of 10 analyses.

THE FIELD · 11 LABS · 129 BATCHES



LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS



HOW RARE

6 of 129 batches outside, in 3 of 11 labs
8 labs never flag

DOES EVERY PLATE AGREE

25 of 25 plates outside · flagged

2 of 194 plates outside · the rest

WHAT CHANGES IT

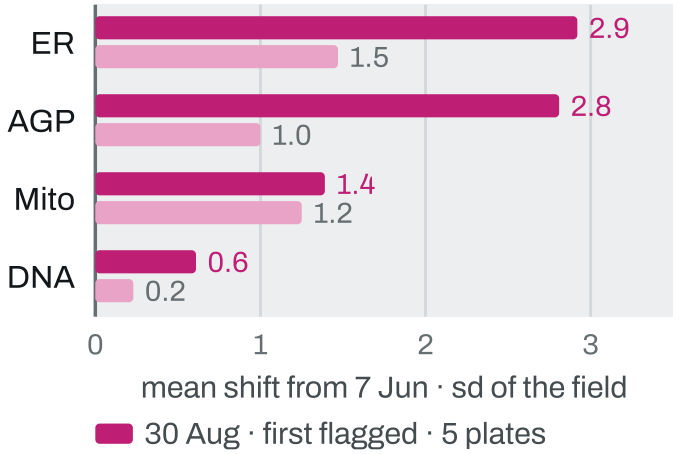
1 of 10 variants differs
its baseline takes in 21 Jun, whose cell count fell

■ in distribution ■ outside ▢ baseline · Outside: further from its own first batch than from the nearest other lab.

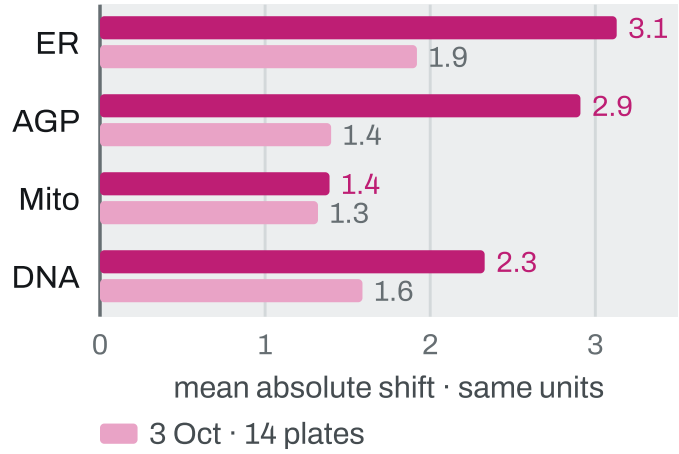
Intensity rose.
Cell count did not.

Lab 2's flagged batches against its first batch, 7 Jun.

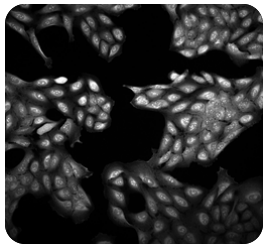
STAIN INTENSITY



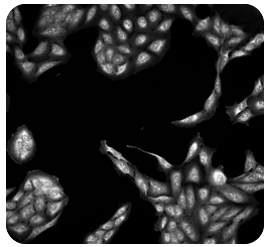
TEXTURE



AS THE MICROSCOPE SHOWS THEM



7 JUN · ER
7,612 counts



30 AUG · ER
51,230 counts

raw counts 7× apart · DNA 20×
pattern: perinuclear → whole cell

READING

96% or more of ER and AGP intensity features rose
cells per well 189 → 165 → 182
inside the lab's own range, 150–199

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition.
Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

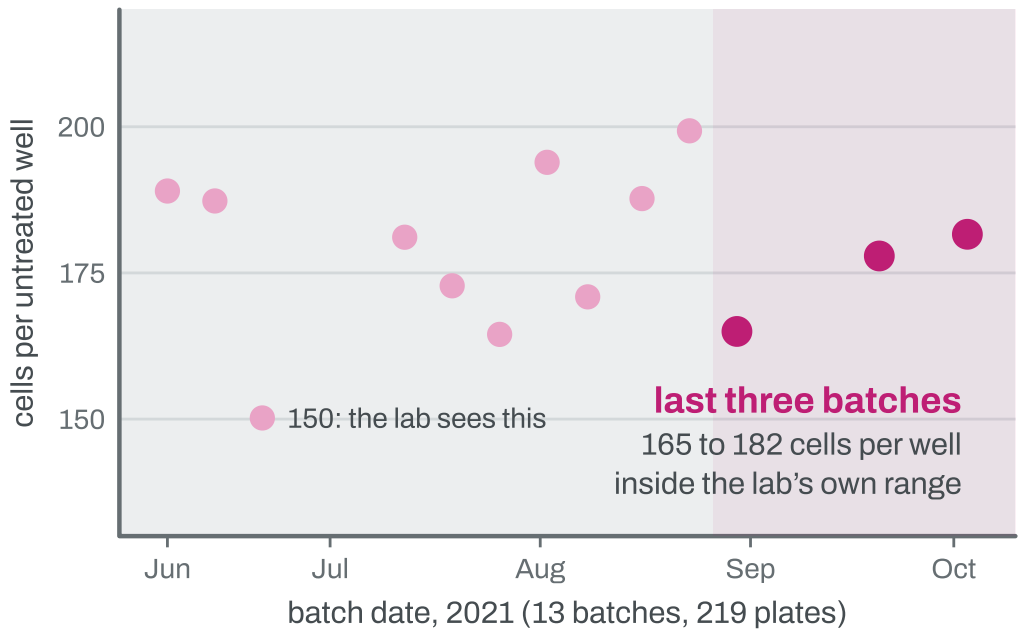
Look first: exposure · ER stain
Check: stain mix-up · bleed-through · biology
Not why: that is the lab's to find

means over the features of one channel · sd: standard deviation of the field · JUMP [7]

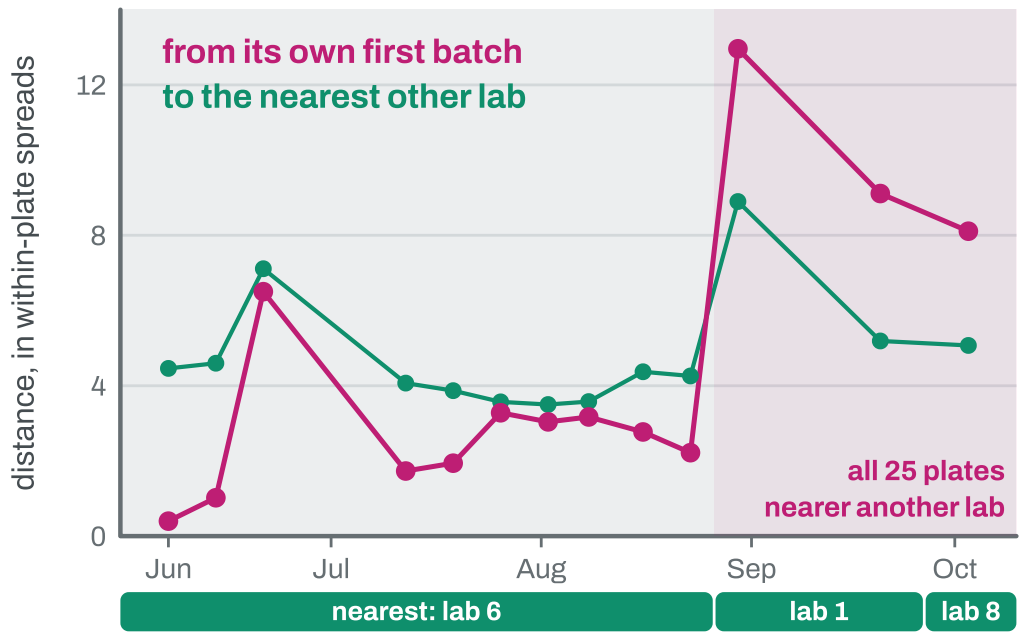
Alone, it looks fine.
Among everyone, it moves.

Same 13 batches, two views.
Not a verdict: a reason to look.

WHAT THE LAB SEES OF ITSELF



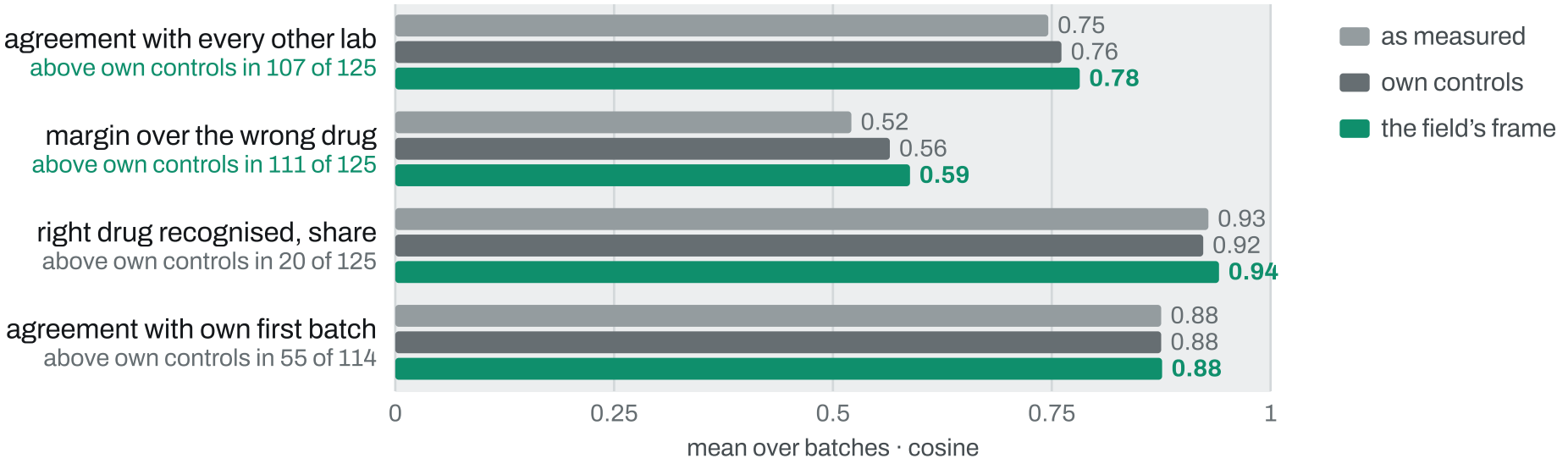
WHERE IT SITS AMONG EVERYONE



unit: the field's within-plate spread · nearest lab: closest centre · lab 2 · U2OS · 2021 · JUMP [7] · what moved: A12

One frame: what it buys.

Untreated wells map each batch onto a field built without its lab.



DOES IT HOLD



batches above own controls · 100–400 components · shrinkage 0.05–0.5

WHAT IT DOES NOT DO

Effect sizes still vary between batches

spread 0.06 as measured · 0.20 own controls · 0.12 field's frame

1,829 plates, 11 labs, 7 positive controls, 200 principal components. Own controls: each feature standardised on the batch's untreated wells.

PROPOSED · PRICES ARE THE ASSUMPTIONS ON A22

Labs contribute free.
Buyers of cell data pay.

The certificate pays first.
The exchange grows from it.

WHO	WHAT THEY GET	WHAT THEY PAY	WHAT IT BUILDS
labs and cores	a place on the map, the certificate, the game	**free, for contributing controls**	the map
companies that buy cell data	a certificate on every dataset they buy	**~$10–50k per certificate**	the revenue
groups that screen continuously	a reading on every batch they run	**~$5–15k per line, per quarter**	the habit
AI labs and model builders	the benchmark of planted failures	**~$50–200k a year, as partners**	the standard
then: the exchange	certified datasets, traded lab to buyer	**a share of each trade; the lab keeps the rest**	the network

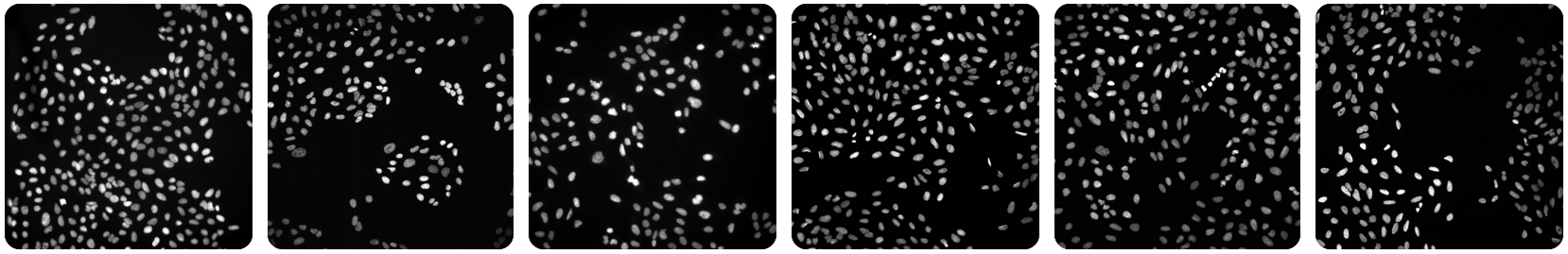
control wells in · readings out · nothing else leaves the lab · each reading is a certificate

Build the ugliest dataset in biology, on purpose.

Same cells, same protocol, no drug.
It already looks like this.

TEST SET · WHAT SIX LABS ALREADY SEND

real wells · JUMP cpg0016 · CC0



LAB 1	LAB 2	LAB 4	LAB 5	LAB 6	LAB 10
Phenix · widefield	CV8000 · confocal	Phenix · widefield	CV8000 · confocal	CV8000 · confocal	CV8000 · confocal

5 microscope models

0.4–1.0 objective NA

0–3 brightfield planes

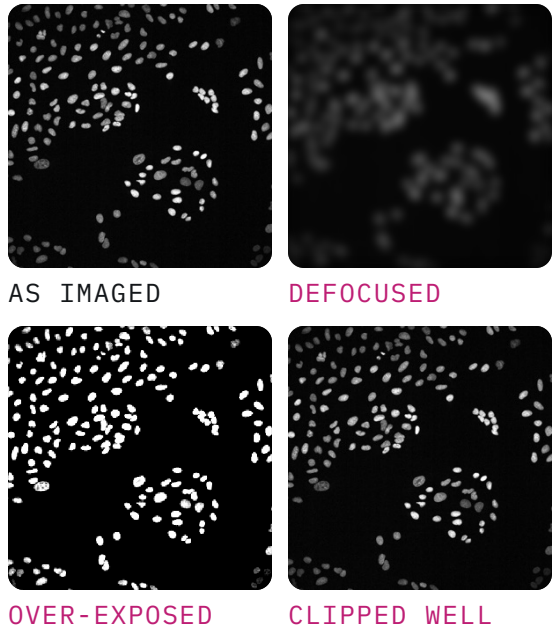
0.60–0.70 µm per pixel

8 plate types

across all 13 JUMP labs · public metadata [7]

RED TEAM

one well · simulated



AS IMAGED	DEFOCUSED	OVER-EXPOSED	CLIPPED WELL

Public benchmark: name the failure.

playground · most-used lines · every microscope · reference perturbations · planted failures · every normalization graded

A game, not a report card.

Every batch ranked on its untreated wells, never on results.

LAB	BATCHES	SCORE	BEST	STREAK	OFF CENTRE	PLATE SPREAD	DRIFT	CELL-COUNT GAP
lab 5	33	**70**	#1	33	3.35	0.27	0.81	0.31
lab 6	13	**60**	#16	13	4.44	0.41	1.23	0.12
lab 10	17	**54**	#10	17	3.84	0.62	1.21	0.23
lab 8	4	**52**	#54	4	4.10	1.08	0.86	0.09
lab 11	5	**47**	#23	2	3.79	0.87	1.57	0.17
lab 3	13	**44**	#18	6	3.27	0.80	2.01	0.29
lab 7	8	**42**	#69	8	6.38	0.58	1.17	0.33
lab 2	13	**39**	#48	10	5.68	0.57	1.99	0.14
lab 1	6	**32**	#78	6	8.68	1.10	2.03	0.09
lab 9	11	**25**	#97	11	9.97	0.50	1.76	0.45
lab 15	6	**23**	#68	6	5.53	0.87	1.69	0.44
field	129	—	—	—	4.01	0.51	1.17	0.28

SCORE

score = 100 − mean percentile

4 measures · among 129 batches · higher is better

BEST BATCH

82

lab 5 · 28 Jun · 7 plates

lab 5 holds 9 of top 10

YOUR CHOICE TO PLAY

Named only if you opt in.

Private version for your own plates.

Lab medians; lower is better on the four measures · 1,871 plates · 11 labs · 129 batches · streak: in distribution, in a row · JUMP [7]

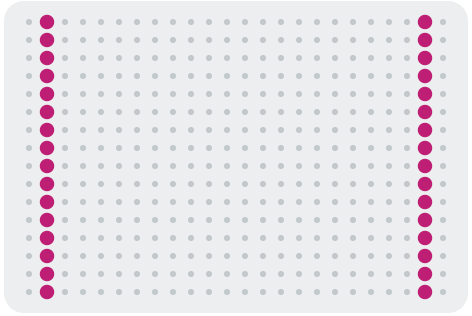
Control wells leave. Nothing else does.

No compound, no target, no result.
Nothing to clear.

WHAT LEAVES THE BUILDING

untreated control wells only
images or well-level profiles, no compound

a plate map of the control wells



pink: 32 of 384 wells leave · JUMP [7]

nothing else

82,319

untreated wells · 10 labs · 0 treated

WHAT DATUM LINE DOES WITH IT

builds the map
one centroid per plate, in a space scaled on every other lab. The lab being placed never sets its own scale (3,773 features here).

answers
position · distance to own baseline · nearest lab · cells per well · verdict · rank

computes a move, not a model of you
one matrix per batch, fitted on your controls · 200 components · A19

WHAT IT NEVER SEES

your treated data
the move is sent to you and applied on your side. Your results are normalised in your building, by you.

your name, unless you choose
proposed: labs appear as a number; a lab opts in to a name. Every JUMP lab here is a number.

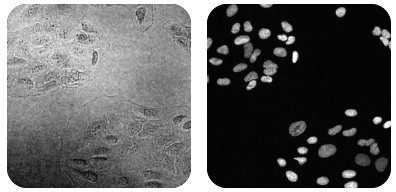
your images, for long
proposed: images are reduced to profiles on arrival and the originals deleted; the profile of a control well is what is kept.

Every analysis here runs on this rule · github.com/kat-titter/datum-line/analysis · JUMP, CC0 [7]

Image in.
Place, distance, move out.

Data stays where it is.
Models and moves travel.

STEP 1 — AN IMAGE



brightfield dna

One untreated well.

STEP 2 — NUMBERS

3,773

numbers per well today

CellProfiler today.
Vision model next:

brightfield 99.6%

dna 97.2%

one picture · names the lab

STEP 3 — THE MAP

99.9%

a classifier names the lab; chance 9%

Z-scored on every other lab; one point per plate.

The lab being placed never sets its own scale.

STEP 4 — THE MOVE

$$x' = C_{field}^{½} C_{batch}^{-½} (x - m_{batch})$$

per batch · 200 components

Your batch onto the field's frame; applied on your side.

One laptop runs the whole replay from cache in minutes. Every script: github.com/kat-titter/datum-line/analysis. What leaves a lab: A18.

Cells cannot lie about where they came from.

One map of untreated wells.
Two checks measured, two proposed.

PROVENANCE

99.9%

of 93,228 untreated wells name their true lab, on plates the classifier never saw.

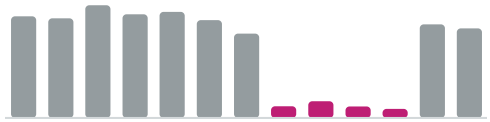


129 batches · 119 every well

INTEGRITY

4 batches

65 plates whose plate map says positive control, and whose cells show no effect; no other well on those plates does either.



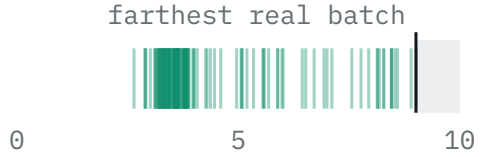
lab 3 · effect per batch · A8

CELLS FROM NOWHERE

9 spreads

the farthest any of 129 real batches sits from every other lab. Beyond that, a culture is not the line it claims, or not alone.

untested · proposed as the first partner-lab test



distance to nearest other lab

SENTINELS

proposed

An untreated well is a canary. If the controls move and nobody changed anything, something in the room did: a lot, a reagent, the air, the water.

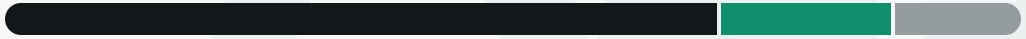


lab 2 · drift by batch · A13

Defensive by construction: untreated wells only · no result or perturbation stored · distances in within-plate spreads · JUMP [7]

About $1.8M for eighteen months.

$884k planned × 2 = $1.77M asked



people 71% · benchmark 17% · running 12%

leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates · planted failures · partner labs or a CRO	**$120k**
partner-lab bench time	3 labs · controls they already image · reagent top-ups · shipping	**$30k**
compute and storage	images · models · hosting · API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners · 3 labs	**$20k**
planned	sum	**$884k**
+ 2× rule	most things take 2–3× · plan at 2×	$884k
asked · 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a CRO.
Own lab: costs more, proves less.

BENCHMARK IS INSIDE

Milestone 3, paid for in this round.
Not a later one.

WHAT IT HAS TO SHOW

1 three labs · **2** any image · **3** playground
4 paid certificate → seed

salaries below market · leader and scientific lead paid · nothing here committed

About $84M a year, if everyone joined.

Count × price.
Three streams.

assumption	low	base	high
organisations screening a few lines deep	200	400	800
lines each runs per quarter	2	4	8
price per line per quarter, $k	5	10	15
organisations screening panels of hundreds to thousands of lines	10	25	50
site licence per year, $k	50	150	300
subscriptions and site licences, per year	**$8M**	**$68M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, $k	10	25	50
certificates on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
serviceable, per year, at full adoption	**$10M**	**$84M**	**$523M**

BASE CASE · PER YEAR



↑ five-year target · a tenth · $8M

benchmark $1M ↑

THE ANCHOR

My own screen, run on cells that had moved: about $2.6M to redo.
Certificate: a slice of that.
Subscription: insurance against it.

Prices are the proposal on A15. Academic labs and cores are free and are not in the revenue rows; they are the map.

Data has a price.
Cell data needs a standard.

Three public prices.
Cell data from four sites does not add up.

AI TRAINING DATA

$1.5M a day

paid to experts for AI training data, by one vendor

~$85 an hour · $10B valuation [10]

CELL MAPS

$30M

one cell map · paid by Roche

second milestone · $213M to date [11]

CONSORTIA

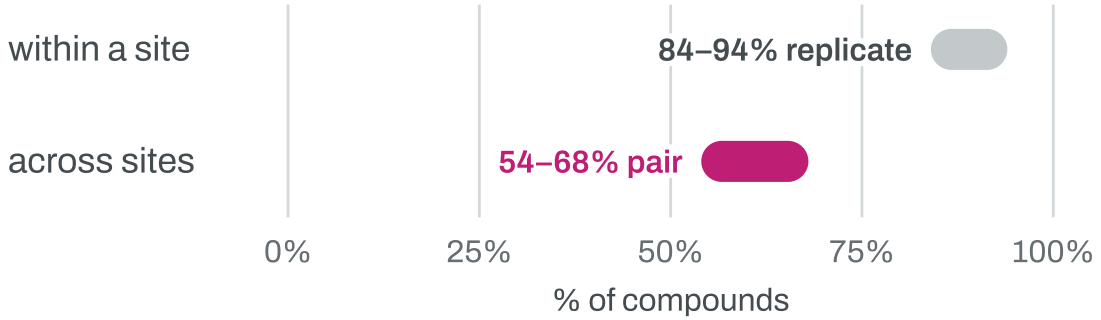
$1.8B

pledged for virtual-cell data; funders get a year first

Meta, DeepMind, Isomorphic: $300M [12]

EU-OPENSCREEN · ONE PROTOCOL · 4 SITES · HEP G2

same cells, serum and dye lots, shipped to every site [13]



ON THE MAP

0.75 → 0.78

labs agree on the same drug · 121 of 125 batches · A14

[10] TechCrunch, 2025 · [11] Recursion, 2025 · [12] eWeek, 2026 · [13] Wolff et al., *iScience*, 2025 · full references in the README

Owned by the lab.
Verified by anyone.

Open science, without the token.

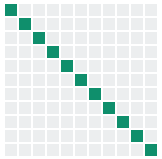
CONTROL WELLS ONLY — today

no compound · no target · no result
frame applied on the lab's side

LABS LEAVE FINGERPRINTS — today

99.9%

wells name their own lab



true lab ↓ · named lab → · held-out plates

SIGNED AND TIMESTAMPED — proposed

public log · anyone checks · nobody backdates
blockchain optional

LAB KEEPS ITS DATA — proposed

listing is opt-in
withdraw any time

KNOWN BUYERS — proposed

verified at sign-up · odd access flagged
access log open to members

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data · no pathogen data

1 **lab lists**
untreated wells
certificate attached

→

2 **certificate checks**
where the cells sit
who made them · what moved

→

3 **buyer pays**
per dataset
per line for screening groups

→

4 **lab keeps a share**
of every sale
as long as the data sells

nothing contracted · privacy A18 · biosecurity A20 · open: map, code, public reference · paid: certificate, A15

PUBLIC SOURCES [6], [7], [9]

Today: search, ask, guess.

WHY NOW

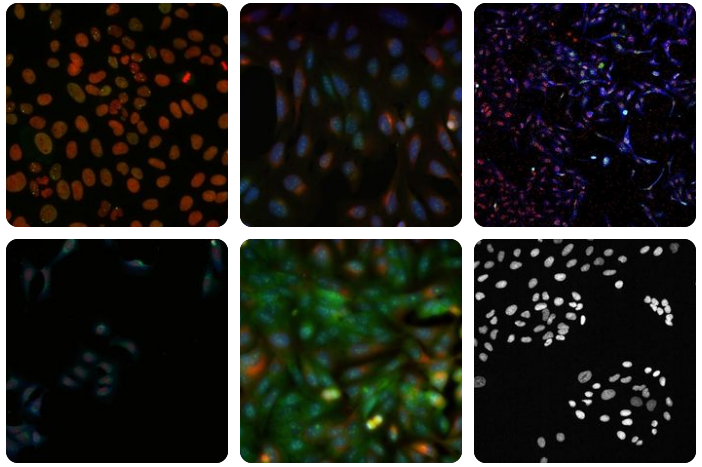
- 2023 — reference goes public [7]
- 2025 — cell data gains regulatory weight [9]
- 2026 — the map

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6] — $28B

36% ≈ $10B · reagents & references | 64% · design, analysis, protocols

1
search

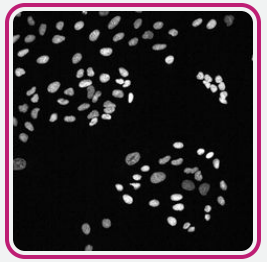
do my U2OS cells look normal



2
ask

To: whoever ran this line last

Subject: does this look right to you?



screenshot.png

3
guess

$ RUN IT ANYWAY · TOO FAST

hundreds of thousands

of R&D

HOLD · TOO SLOW

months

of development

[6] Freedman 2015 · six real U2OS papers